Design Details Document

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Design of Robot 1 (R1):

Overall dimensions (in mm) and estimated weight (in Kgs):

Length	682mm
Width	686mm
Height	863mm
Weight	23.2kg(appx.)

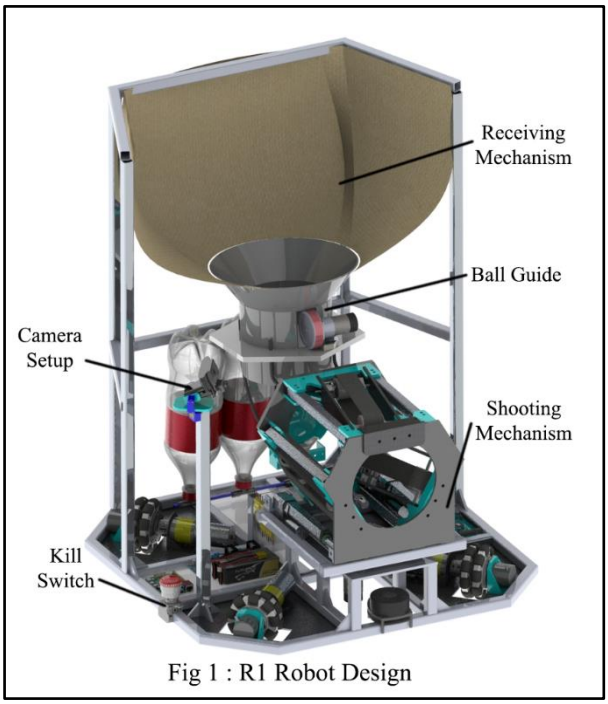
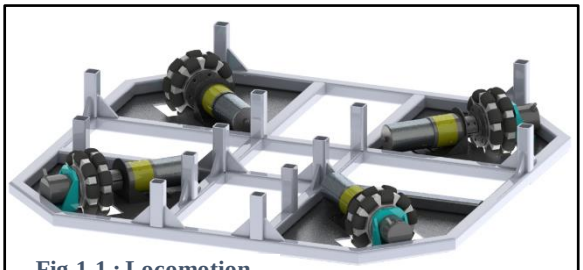


Fig 1 : R1 Robot Design

Type of Drive (4-wheel holonomic Omni drive):

The holonomic cross drive with Omni wheels is selected for its capability to move to any coordinate in the X-Y plane by calculating the instantaneous resultant of all 4 wheels, while providing control over the rotation about the Z-axis as well. Thus, the Holonomic Omni cross drive has an advantage in terms of manoeuvrability and traction.

The Seeker Robot R1 chassis is made up of a 6061-grade Aluminium tube (20mm x 20mm x 2mm) which is lightweight, structurally rigid, and ensures sturdiness. The locomotion of R1 is semi-autonomous. It can be completely controlled by a manual operator using a PS2 controller and can also be programmed to reach a specified coordinate by pressing a button on the controller. Integration of RP LiDAR with Sensor fusion of encoders and IMU is used to make the locomotion of R2 autonomous and is flexible enough to be switched to manual control, making R2 semi-autonomous. Feedback data from these sensors is sent to ROS (Robot Operating System), the software running on a Jetson Nano, which processes and issues suitable commands to the Arduino Due. Subsequently, Arduino sends appropriate signals to the motor drivers that control the locomotion of the robot.


Fig 1.1 : Locomotion

Ball receiving mechanism:

The ball thrown by R2 is received by R1 into a funnel-like structure covered with nylon mesh. The mesh helps to damp the momentum of the ball and ensures safe receiving.

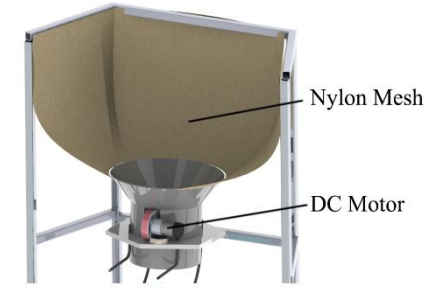


Fig 1.2 : Receiving Mechanism

Brief description of Lagori breaking (Seeker R1):

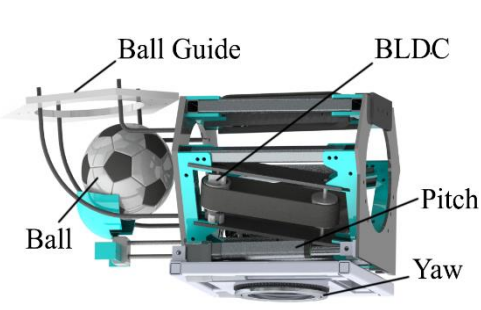


Fig 1.3 : Shooting Mechanism

To load the ball into the shooter for lagori breaking, 3 balls are stacked in the receiving funnel of R1. At the end of the funnel all the balls pass through the reducer (DC motor mounted on it) so that the shooter is loaded with one ball at a time.

The shooting mechanism consists of a shooter enclosed in a cylindrical frame made with aluminium tubes and sheets. The shooter comprises of 3 belts, each running along a driven pulley connected to a BLDC motor and an Idler pulley. These belts allow

the ball to travel effectively at a long range without diversion from the projected path.
To break the lagori pile for the first try, the ball is targeted at the bottom most disk with a velocity of 7.4m/s and angle of launch as 18.1° to overcome the friction between the disk and the lagori base, there by displacing the disks by around 150mm. In the second try the ball is launched with the velocity of 8.1m/s and angle of launch as 14.75, displacing the bottom disk completely from the lagori base. Thus, breaking the pile completely.
To achieve the desired projectile for each, try, the yaw and pitch of the shooter are adjusted with the help of a planetary geared motor and linear actuators respectively. The required angle of launch is derived by considering the launch velocity of the ball varying between 8.1m/s to 8.3m/s, using the projectile equations mentioned below.

$$W = F \times D \quad \text{.....1}$$

$$W = \frac{1}{2}mv^2 \quad \text{.....2}$$

From 1&2

$$v_2 = \sqrt{\frac{2FD}{m}} \quad \text{.....3}$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \quad \text{.....4}$$

$$u_1 + v_1 = u_2 + v_2 \quad \text{.....5}$$

$(u_2 = 0)$

From 4 & 5

$$u_1 = 8.1m/s$$

Where w = work done to displace lagori, F = Force required to displace lagori.
D = Displacement of lagori, m = mass of all lagoris.
v_2 = Final velocity of lagori, v_1 = Final velocity of ball.
u_1 = Initial velocity of ball, u_2 = Initial velocity of lagori.

$$x = u\cos\theta.t \Rightarrow t = \frac{x}{(u\cos\theta)}$$

$$\Rightarrow y - y_0 = u\sin\theta.t - \frac{1}{2}gt^2$$

$$\Rightarrow y - y_0 = u\sin\theta.\frac{x}{(u\cos\theta)} - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2$$

$$\Rightarrow y - y_0 = u\sin\theta.\frac{x}{(u\cos\theta)} - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2$$

$$\Rightarrow y - y_0 = x.\tan\theta - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2$$

$$\theta = \frac{\tan^{-1}\left(\frac{y-y_0}{x}\right) + \sin^{-1}\left(\frac{gx^2 + (y-y_0)v^2}{v^2\sqrt{x^2 + (y-y_0)^2}}\right)}{2}$$

$$\theta = 14.75^0$$

where θ = Launching Angle,
y=height of the destination pot, y_0= launching height,
g= acceleration due to gravity, x=Horizontal Range
t=time of flight, u=launching velocity

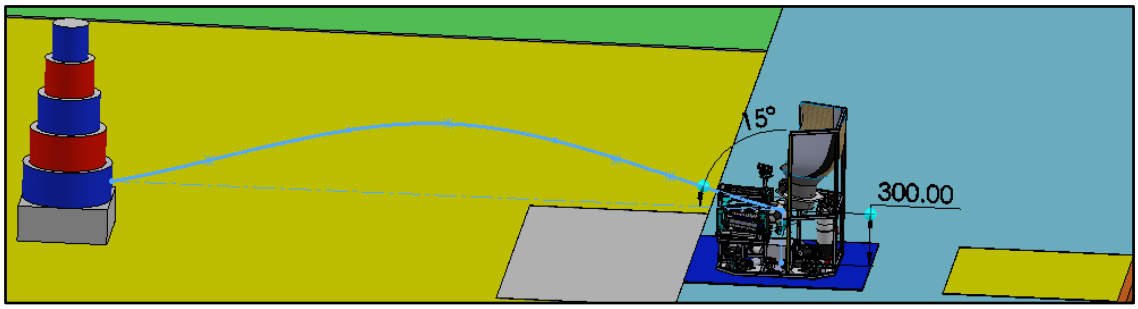


Fig 1.4 : Breaking Lagori

Design of Robot 2 (R2):

Overall dimensions (in mm) and estimated weight (in Kgs):

Length	888mm
Width	712mm
Height	1200mm(with ball) 691(without ball)
Weight	20.8kg

Type of Drive (4-wheel Swerve drive):

The Robot R2 employs Swerve drive which has superior manoeuvring capabilities over other drives and is mounted onto a chassis made of 20 x 20 x 2 mm 6061 grade Aluminium square tubes. At each corner of chassis, a swerve drive setup(Fig 2.1) is used, in which a BLDC motor is the source of rotation of the wheel as it has low noise level, great durability and relatively high amount of

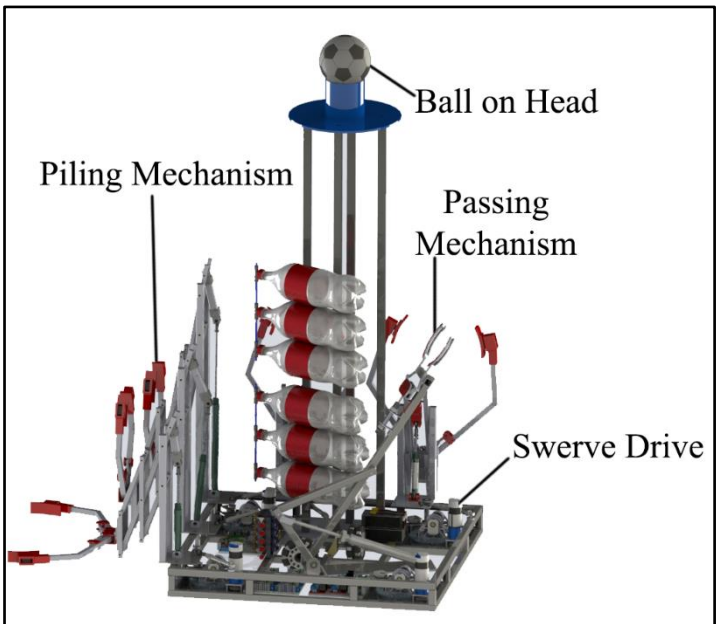


Fig 2 : R2 Robot Design

torque transmission, while a planetary geared motor is used to steer the wheel.
During translation of the Robot, each wheel is powered individually to propel the robot and is directed accurately by the planetary geared motor. This motor needs to rotate in any direction in order to control the orientation of the robot. The power is transmitted with the help of gears. Integration of RP LiDAR with Sensor fusion of encoders and IMU is used to make the locomotion of R2 autonomous and is flexible enough to be switched to manual control, making R2 semi-autonomous . Two omni wheels(acting as dummy wheels) are attached perpendicular to each other and optical encoders are attached to them for getting precise feedback of locomotion. From the above discussion, this drive has an advantage in terms of speed compared to the other drives. Thus the Swerve drive mechanism is selected due to its precise working, ease of operation, speed, acceleration and efficiency.

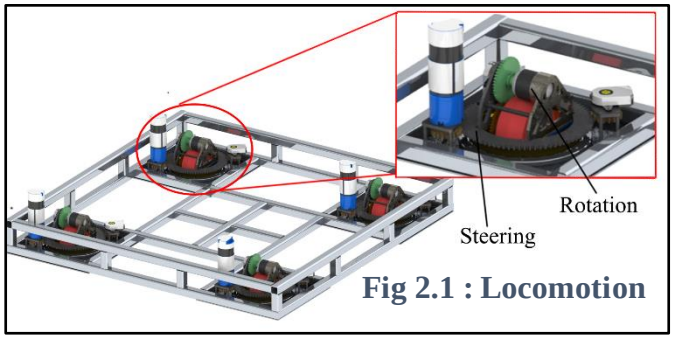
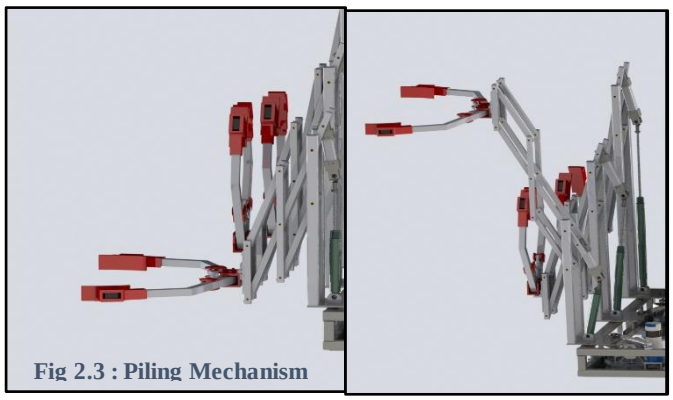


Fig 2.1 : Locomotion

Piling up the Lagori discs:
The camera mounted on the robot R2 detects the broken lagori discs in the Lagori area and align itself to pick them up. The servos arebattached to the camera for better view. The angle of the camera adjusts itself as it approaches the disc. The disc size is recognised by the camera as it is trained with a dataset of piling disc images. the system then sends the signal to the respective dedicated sized grabber to grab that disc.
A six-bar mechanism made of aluminium tubes is used to pick the lagori disc up. The 10x10 aluminium tubes are connected in such a way that when pivoted from one end in the middle, the link of the other end which is connected to the claw stays in the same orientation, thus allowing the lagori pile to remain uninclined. The whole mechanism is actuated by a pneumatic piston, allowing fast translation of a lagori pile to the desired height. The end effector consists of a pneumatic claw mechanism with two fingers. Each finger has a servo installed inside it, allowing them to change the orientation of the lagori pile if needed. The open & close of this mechanism is also achieved by pneumatic pistons placed on each finger. This mechanism helps us in achieving the pick-and-pile task with high speed & high rigidity. The easily expandable feature of the six-bar mechanism deals with the height constraints, as it balloons out exponentially to its initial length & height. The Robot R2 is installed with 5 such mechanisms to employ independent handling of lagori piles with varying diameters. Once all the lagori piles are picked, the 5 claws are aligned in such a way that the lagori piles are concentric and above each other. This alignment is achieved by a separate group of pneumatic pistons which pivot the 5 mechanisms about the vertical axis. The lagori piles are then released onto the Lagori base at an instant, giving the robot a fast mechanism and achievement of the lagori pile.


Fig 2.3 : Piling Mechanism

Brief description of the mechanism of Hitter R1 of Hitting of Ball on Head (BOH):
There is a common shooting mechanism for Lagori breaking and shooting the BOH. In the case of shooting BOH, a camera setup is used as shown in fig 2.3. Using the Yolov5 machine learning algorithm, the camera placed on the Hitter R1 detects the ball on the opponent Seeker R2's head and aims it with the centre of the camera frame as a reference. The camera setup has 2 degrees of freedom and is controlled using the servos attached to the camera module .To ensure that the camera is aiming at the ball, a laser light is placed parallel above the camera .The distance of the ball from the bot is estimated using a Lidar placed below the camera as shown in fig 2.3. Once the distance is estimated, the required yaw and pitch values for shooting the BOH are calculated using the projectile equations. Hence the shooter is adjusted and the ball is launched.

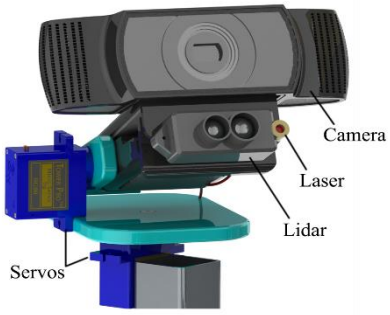


Fig 2.3 : Camera Setup



Fig 2.4 : Pitch of Shooting Mechanism

Ball picking and passing mechanism (Hitter R2):

To pick balls from the Ball rack, the robot R2 approaches it and orients itself accordingly with the help of odometry and an RP Lidar. The picking mechanism is a pneumatic catapult with a thin claw as its end effector. One end of the aluminium catapult has the claw and the other end is pivoted to the chassis. Another connection of the aluminium tube with the chassis is the pneumatic piston which actuates it. The piston is connected in such a way that, if the stroke length of the aluminium tube increases, the end effector gains angular velocity and makes a curve path with high speed, to achieve the task of throwing the ball to the robot R1. Both the Catapult and the claw are actuated by pneumatic pistons. The robot R2 aligns the claw concentric to the ball in its open position. The claw is then closed to hold the ball from its bottom. The closed radius of the claw has a diameter less than that of the ball. The robot then catapults the ball to the robot R1. The ball is released from the claw without it being opened as the ball is just resting on it. The robot R2 has a type-B laser, which is fixed at the frontal part of the robot in order to align it with R1 before the pass. When the robots are aligned, the piston actuates, and passes the ball to the robot R2 with a velocity of 8.3 m/s at an angle of 45° .

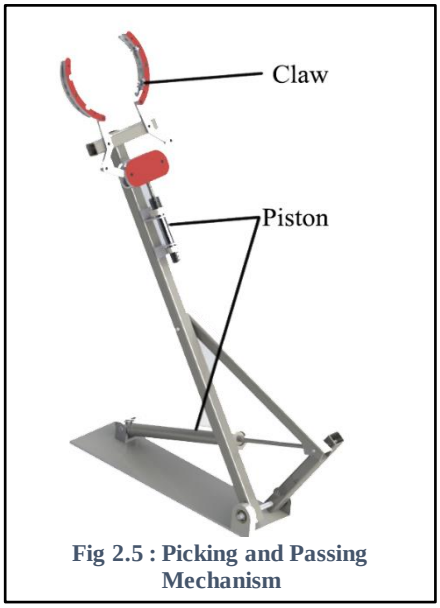


Fig 2.5 : Picking and Passing Mechanism

$$x = u\cos\theta \,.\, t \quad \text{------>1.}$$

$$t = \frac{x}{u\cos\theta} \quad \text{------>2.}$$

$$y = u\sin\theta \,.\, t - \frac{1}{2}gt^2 \quad \text{------>3.}$$

$$y = u\sin\theta \,.\, \frac{x}{u\cos\theta} - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2 \quad \text{------>4.}$$

$$y = x.\tan\theta - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2 \quad \text{------>5.}$$

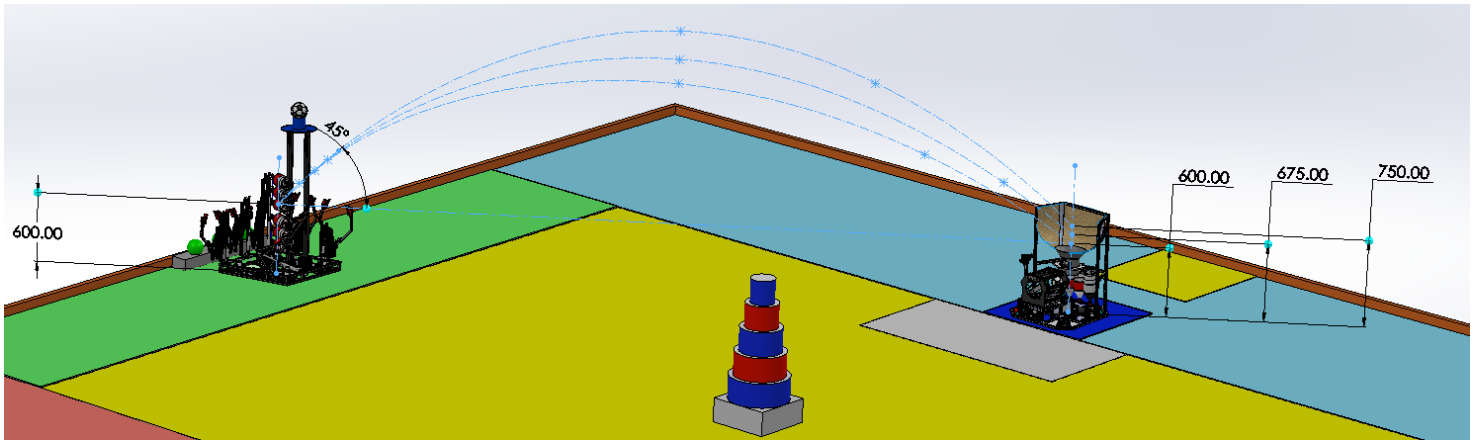


Fig 2.6 : Simulation of passing

Type of sensors integrated:

S.no	Component	Specifications	Usage
1	Optical Encoder	Incremental type, 3.3V-5V DC, PPR=360, CPR=1440.	(R1): Attached to Omni wheels to monitor the distance travelled by the bot.
2	Through Bore Encoder	Incremental type, 3.8V-5V DC, 48 to 2048 PPR adjustable by a DIP switch.	(R2): Attached on PU wheels in swerve-drive mechanism to monitor the rotation of the wheel, and the distance travelled by the bot.
3	Through Bore Encoder	Absolute type, 3.8V-5V DC, with 14-bit multi-turn absolute resolution.	(R2): To monitor the steering of the wheel for precise turning of the bot.
4	RP LiDAR	5-10V DC, with 360°2-D Laser Scanner, with a maximum range of 12m.	(R1, R2): To map the whole arena to achieve automation and to avoid obstacles.
5	Logitech C920 Camera	4.5-6V DC, with a maximum range of 12m, with 78° View Angle	(R1, R2): For Object detection
6	TF mini LiDAR	3.3V-5V DC, Maximum range of 12m, with a 4° Field of View	(R1, R2): To determine the distance from the lagori discs and BOH to adjust the shooting mechanism.
7	IMU MPU- 9250	3.3V-5V DC, 9-axis gyroscope,	(R1, R2): To determine the acceleration and position of the bot.
8	IR Sensor	5V DC,10-20 cm.	(R1): To control the ball flow from receiver to the shooter.

Type of Actuators integrated:

S.no	Component	Specifications	Required power	Usage
1	Planetary geared motor	450 RPM, 24V DC, 8kg-cm (stall torque),	38W	Locomotion (R1): 4 motors attached to the wheels
2	DC geared motor	6000 RPM 24V 7kg-cm (Stall Torque)	30W	Steering (R2): 4 motors are used for independent steering of the wheel.
3	Servo Motor R1-MG996R R2-MG90S	5V DC, 10kg-cm (Stall Torque)	10W	(R1): To control the yaw and pitch angles of the camera for object tracking. (R2): To rotate the lagori discs if it falls perpendicular to the ground.
4	BLDC Motor (for R1)	22.2V (6s) DC, 20.5 kg-cm(Stall Torque)	800W (max)	(R1): For shooting balls to displace lagori discs and opponent's BOH.
5	BLDC Motor (for R2)	11.1V (3s) DC, 15 kg-cm(Stall Torque)	620W (max)	(R2): For locomotion which drives each wheel in swerve-drive mechanism.
6	Pneumatic Piston with solenoid	12V DC, 25mm,200mm & 420mm stroke length	1.2 W	(R2):To pick and lift the lagori piles in 6 bar mechanism and to grab and pass the balls from rack.
7	Linear actuator	12V DC,6mm/sec, 100mm stroke length	25W (max)	(R1): To control the pitch angle of the shooting mechanism.
8	Planetary geared motor	300 RPM, 12V DC, 50 kg-cm (stall torque)	40W (max)	(R1)To control the yaw angle of the shooting mechanism.
9	Relay	12V DC	—	(R1, R2): To manage battery levels.
10	DC geared motor	60RPM, 12V DC, 25kg-cm(Stall torque)	20W (max)	(R1): To send one ball at a time in the shooter.